

RQ Answers

omitted

2024-04-05

Data overview

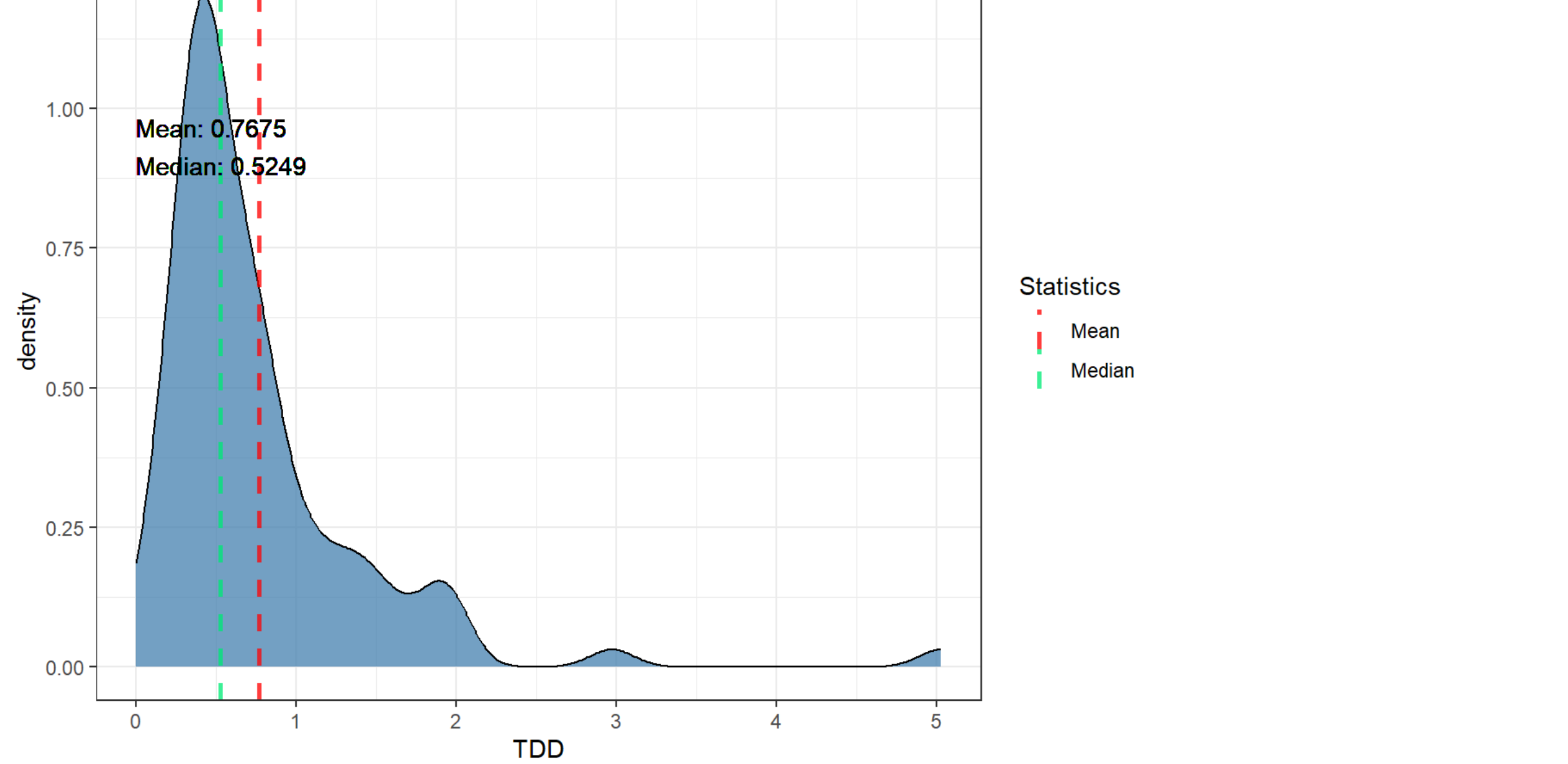
glimpse(all_data)	
## Rows: 4,140	
## Columns: 11	
## \$ repo	<chr> "struts", "struts", "struts", "struts", "struts", ...
## \$ pr_number	<int> 286, 286, 286, 286, 286, 286, 286, 286, 286, ...
## \$ rule	<chr> "java:S1192", "java:S125", "java:S1192", "java:S11...
## \$ severity	<chr> "CRITICAL", "MAJOR", "CRITICAL", "CRITICAL", "CRIT...
## \$ file	<chr> "core/snc/test/java/org/apache/struts2/views/jsp/u...
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SML...
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...
## \$ debt	<int> 8, 5, 18, 18, 12, 18, 10, 18, 12, 5, 5, 5, 5, 5...
## \$ ncloc_affected_file	<int> 209, 209, 209, 209, 209, 209, 209, 209, 209, ...
## \$ complexity	<int> 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, ...
## \$ origin	<chr> "NEW", "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE...

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd)), : All aesthetics have length 1, but the data has 98 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd)), : All aesthetics have length 1, but the data has 98 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

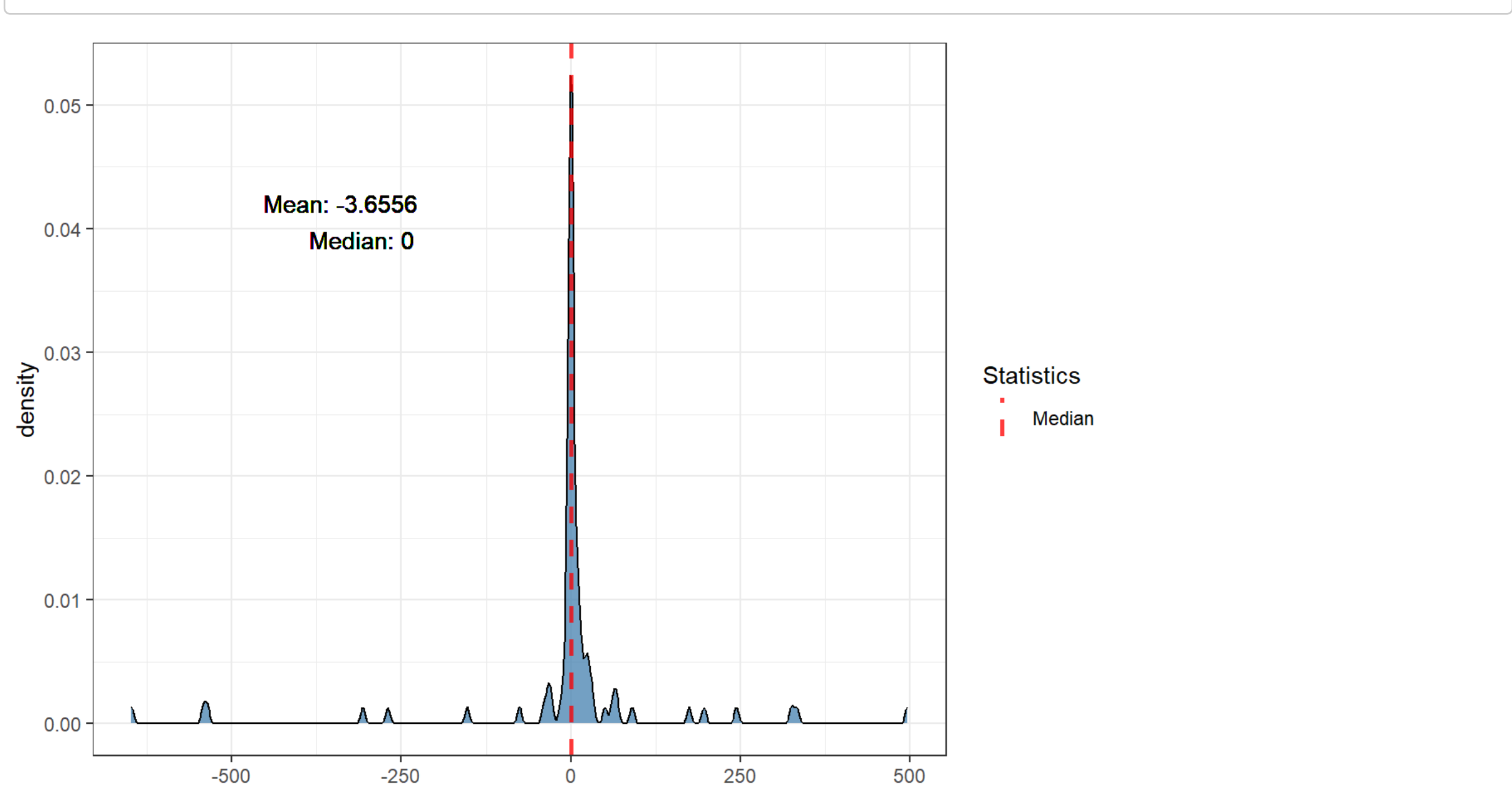


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv)), : All aesthetics have length 1, but the data has 98 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv)), : All aesthetics have length 1, but the data has 98 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 struts	26.7	36.7	36.7	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	
## 1	26.7	36.7	36.7

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 struts	56.5	43.5	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	56.5	43.5

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 struts	97.9	2.08	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	97.9	2.08

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a table of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank struts		
## <int> <chr>		
## 1	1 java:S3740	- 116
## 2	2 java:S1192	- 82
## 3	3 java:S1874	- 71
## 4	4 java:S2293	- 24
## 5	5 java:S1171	- 15
## 6	6 java:S3599	- 15
## 7	7 java:S1186	- 14
## 8	8 java:S1161	- 11
## 9	9 java:S3252	- 9
## 10	10 java:S112	- 6

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S3740	122	
## 2 java:S1192	123	
## 3 java:S1874	124	
## 4 java:S2293	125	
## 5 java:S1171	126	
## 6 java:S3599	127	
## 7 java:S1186	128	
## 8 java:S1161	129	
## 9 java:S3252	130	
## 10 java:S112	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank struts		
## <int> <chr>		
## 1	1 java:S1192	- 1286
## 2	2 java:S3740	- 319
## 3	3 java:S1874	- 293
## 4	4 java:S117	- 245
## 5	5 java:S3599	- 178
## 6	6 java:S1171	- 178
## 7	7 java:S2293	- 116
## 8	8 java:S2129	- 104
## 9	9 java:S2259	- 90
## 10	10 java:S1186	- 71

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1192	122	
## 2 java:S3740	123	
## 3 java:S1874	124	
## 4 java:S117	125	
## 5 java:S1171	126	
## 6 java:S3599	127	
## 7 java:S2293	128	
## 8 java:S2129	129	
## 9 java:S2259	130	
## 10 java:S1186	131	

Outliers

Higher TDD

## # A tibble: 98 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 struts	807	196	39	5.03
## 2 struts	594	300	101	2.97
## 3 struts	670	604	302	2.
## 4 struts	607	1603	822	1.95
## 5 struts	559	1603	825	1.94
## 6 struts	417	782	414	1.89
## 7 struts	313	1080	597	1.81
## 8 struts	765	1378	817	1.69
## 9 struts	666	927	598	1.55
## 10 struts	526	1073	693	1.55
## # i 80 more rows				

Higher TDV

## # A tibble: 98 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	559 struts	496	
## 2	427 struts	333	
## 3	417 struts	325	
## 4	861 struts	244	
## 5	807 struts	196	
## 6	612 struts	174	
## 7	526 struts	90	
## 8	611 struts	68	
## 9	549 struts	65	
## 10	307 struts	61	
## # i 80 more rows			

Lower TDV

## # A tibble: 98 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	799 struts	-648	
## 2	670 struts	-542	
## 3	826 struts	-536	
## 4	631 struts	-307	
## 5	622 struts	-270	
## 6	625 struts	-153	
## 7	664 struts	-76	
## 8	841 struts	-40	
## 9	666 struts	-35	
## 10	577 struts	-31	
## # i 80 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 85	

Number of java:S1192 issues that affect test code

## n	
## 1 1368	

Number of java:S1192 issues that affect not test code

## n	
## 1 0	

Summary NCLOC

## ncloc	
## Min. : 12.0	
## 1st Qu.: 173.8	
## Median : 430.5	
## Mean : 655.2	
## 3rd Qu.: 923.8	
## Max. : 4149.0	